Question 1: Find the eigenvalues and eigenvectors of each matrix **(8 marks)**

(a) $\begin{bmatrix} -1 & 6 \\ -1 & 4 \end{bmatrix}$

(b) $\begin{bmatrix} -1 & 1 \\ -8 & 5 \end{bmatrix}$

(a) $\det \begin{bmatrix} -1-\lambda & 6 \\ -1 & 4-\lambda \end{bmatrix} = 0$

$\therefore (-1-\lambda)(4-\lambda) + 6 = \lambda^2 - 3\lambda + 2 = (\lambda - 1)(\lambda - 2) = 0$

$\therefore \lambda_1 = 1 , \lambda_2 = 2$

For $\lambda_1 = 1$,

$\begin{bmatrix} -2 & 6 \\ -1 & 3 \end{bmatrix} u_1 = 0 \quad \therefore \begin{bmatrix} -2 & 6 \\ 0 & 0 \end{bmatrix} u_1 = 0 \quad \therefore u_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$

For $\lambda_2 = -2$,

$\begin{bmatrix} -3 & 6 \\ -1 & 2 \end{bmatrix} u_2 = 0 \quad \therefore \begin{bmatrix} -3 & 6 \\ 0 & 0 \end{bmatrix} u_2 = 0 \quad \therefore u_2 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

Question 1: Find the eigenvalues and eigenvectors of each matrix **(8 marks)**

(a) $\begin{bmatrix} -1 & 6 \\ -1 & 4 \end{bmatrix}$ (b) $\begin{bmatrix} -1 & 1 \\ -8 & 5 \end{bmatrix}$

(b) $\det \begin{bmatrix} -1-\lambda & 1 \\ -8 & 5-\lambda \end{bmatrix} = 0$

$\therefore (-1-\lambda)(5-\lambda)+8 = \lambda^2 - 4\lambda + 3 = (\lambda-1)(\lambda-3) = 0$

$\therefore \lambda_1 = 1, \lambda_2 = 3$

For $\lambda_1 = 1$

$\begin{bmatrix} -2 & 1 \\ -8 & 4 \end{bmatrix} u_1 = 0 \quad \therefore \begin{bmatrix} -2 & 1 \\ 0 & 0 \end{bmatrix} u_1 = 0 \quad \therefore u_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

For $\lambda_2 = 3$,

$\begin{bmatrix} -4 & 1 \\ -8 & 2 \end{bmatrix} u_2 = 0 \quad \therefore \begin{bmatrix} -4 & 1 \\ 0 & 0 \end{bmatrix} u_2 = 0 \quad \therefore u_2 = \begin{bmatrix} 1 \\ 4 \end{bmatrix}$

Question 2: Find the LU decomposition of each matrix, and hence find the determinant of each matrix **(8 marks)**

(a) $\begin{bmatrix} 2 & 3 & 1 \\ 6 & 13 & 7 \\ 8 & 36 & 29 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & 2 & 0 & 0 \\ 3 & 6 & 4 & 9 \\ 2 & 6 & 3 & 9 \end{bmatrix}$

(a) $$A = \begin{bmatrix} 2 & 3 & 1 \\ 6 & 13 & 7 \\ 8 & 36 & 29 \end{bmatrix} = LU = \begin{bmatrix} & & \\ 3 & & \\ 4 & & \end{bmatrix}\begin{bmatrix} 2 & 3 & 1 \\ 0 & 4 & 4 \\ 0 & 24 & 25 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 4 & 6 & 1 \end{bmatrix}\begin{bmatrix} 2 & 3 & 1 \\ 0 & 4 & 4 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore \det(A) = \det(LU) = \det(L)\det(U) = 1 \cdot 8 = 8$$

(b) $$B = \begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & 2 & 0 & 0 \\ 3 & 6 & 4 & 9 \\ 2 & 6 & 3 & 9 \end{bmatrix} = LU = \begin{bmatrix} & & & \\ 0 & & & \\ 3 & & & \\ 2 & & & \end{bmatrix}\begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 2 & 1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} & & & \\ 0 & & & \\ 3 & 0 & & \\ 2 & 1 & & \end{bmatrix}\begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 3 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 3 & 0 & 1 & 0 \\ 2 & 1 & 1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 & 1 & 3 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

$$\therefore \det(B) = \det(LU) = \det(L)\det(U) = 1 \cdot 6 = 6$$

Question 3: Find the determinant of each matrix using a Laplace expansion **(8 marks)**

(a) $\begin{bmatrix} 0 & 1 & 0 \\ 2 & 1 & 4 \\ 1 & 7 & 3 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 0 & 4 & 1 \\ 2 & 0 & 3 & 0 \\ 1 & 3 & 0 & 0 \\ 4 & 0 & 1 & 2 \end{bmatrix}$

(a) $\det = 0 - 1 \begin{vmatrix} 2 & 4 \\ 1 & 3 \end{vmatrix} + 0 = -1 \cdot (6-4) = -2$

(b) $\det = -0 + 0 - 3 \begin{vmatrix} 1 & 4 & 1 \\ 2 & 3 & 0 \\ 4 & 1 & 2 \end{vmatrix} + 0 = -3 \left(-2 \begin{vmatrix} 4 & 1 \\ 1 & 2 \end{vmatrix} + 3 \begin{vmatrix} 1 & 1 \\ 4 & 2 \end{vmatrix} - 0 \right)$

$= -3(-2 \times 7 + 3 \times (-2)) = -3 \times (-20) = 60$

Question 4: Solve each linear system using Cramer's rule **(8 marks)**

(a) $\begin{bmatrix} 3 & 2 & 1 \\ 0 & 2 & 1 \\ 0 & 4 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 5 \\ 11 \end{bmatrix}$

(b) $\begin{bmatrix} 2 & 1 & 1 \\ 3 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 12 \\ 4 \end{bmatrix}$

(a) $\det \begin{bmatrix} 3 & 2 & 1 \\ 0 & 2 & 1 \\ 0 & 4 & 3 \end{bmatrix} = 3 \begin{vmatrix} 2 & 1 \\ 4 & 3 \end{vmatrix} - 0 + 0 = 3 \times 2 = 6$

$$\therefore \quad x_1 = \frac{\det \begin{bmatrix} 8 & 2 & 1 \\ 5 & 2 & 1 \\ 11 & 4 & 3 \end{bmatrix}}{6} = \frac{-5 \begin{vmatrix} 2 & 1 \\ 4 & 3 \end{vmatrix} + 2 \begin{vmatrix} 8 & 1 \\ 11 & 3 \end{vmatrix} - 1 \begin{vmatrix} 8 & 2 \\ 11 & 4 \end{vmatrix}}{6}$$

$$= \frac{-5 \times 2 + 2 \times 13 - 10}{6} = \frac{6}{6} = 1$$

$$x_2 = \frac{\det \begin{bmatrix} 3 & 8 & 1 \\ 0 & 5 & 1 \\ 0 & 11 & 3 \end{bmatrix}}{6} = \frac{3 \begin{vmatrix} 5 & 1 \\ 11 & 3 \end{vmatrix} - 0 + 0}{6} = \frac{3 \times 4}{6} = 2$$

$$x_3 = \frac{\det \begin{bmatrix} 3 & 2 & 8 \\ 0 & 2 & 5 \\ 0 & 4 & 11 \end{bmatrix}}{6} = \frac{3 \begin{vmatrix} 2 & 5 \\ 4 & 11 \end{vmatrix} - 0 + 0}{6} = \frac{3 \times 2}{6} = 1$$

$$\therefore \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

Question 4: Solve each linear system using Cramer's rule **(8 marks)**

(a) $\begin{bmatrix} 3 & 2 & 1 \\ 0 & 2 & 1 \\ 0 & 4 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 5 \\ 11 \end{bmatrix}$ (b) $\begin{bmatrix} 2 & 1 & 1 \\ 3 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 10 \\ 12 \\ 4 \end{bmatrix}$

(b) $\det \begin{bmatrix} 2 & 1 & 1 \\ 3 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} = 1 \begin{vmatrix} 1 & 1 \\ 1 & 0 \end{vmatrix} - 0 + 1 \begin{vmatrix} 2 & 1 \\ 3 & 1 \end{vmatrix} = -1 - 1 = -2$

$$\therefore x_1 = \frac{\det \begin{bmatrix} 10 & 1 & 1 \\ 12 & 1 & 0 \\ 4 & 0 & 1 \end{bmatrix}}{-2} = \frac{1 \begin{vmatrix} 12 & 1 \\ 4 & 0 \end{vmatrix} - 0 + 1 \begin{vmatrix} 10 & 1 \\ 12 & 1 \end{vmatrix}}{-2} = \frac{-4-2}{-2} = 3$$

$$x_2 = \frac{\det \begin{bmatrix} 2 & 10 & 1 \\ 3 & 12 & 0 \\ 1 & 4 & 1 \end{bmatrix}}{-2} = \frac{1 \begin{vmatrix} 3 & 12 \\ 1 & 4 \end{vmatrix} - 0 + 1 \begin{vmatrix} 2 & 10 \\ 3 & 12 \end{vmatrix}}{-2} = \frac{-6}{-2} = 3$$

$$x_3 = \frac{\det \begin{bmatrix} 2 & 1 & 10 \\ 3 & 1 & 12 \\ 1 & 0 & 4 \end{bmatrix}}{-2} = \frac{-1 \begin{vmatrix} 3 & 12 \\ 1 & 4 \end{vmatrix} + 1 \begin{vmatrix} 2 & 10 \\ 1 & 4 \end{vmatrix} - 0}{-2} = \frac{-2}{-2} = 1$$

$$\therefore \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 3 \\ 1 \end{bmatrix}$$

Question 5: Find the null space and nullity of each matrix **(8 marks)**

(a) $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 1 & 3 & 2 \\ 2 & 3 & 6 & 1 \\ 2 & 2 & 6 & 4 \end{bmatrix}$

(a) $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$

$\therefore \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$\therefore \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

$\therefore N = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

$\therefore$ Nullity $= 1$

Question 5: Find the null space and nullity of each matrix **(8 marks)**

(a) $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & 1 \\ 1 & 3 & 2 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 1 & 3 & 2 \\ 2 & 3 & 6 & 1 \\ 2 & 2 & 6 & 4 \end{bmatrix}$

(b) $\begin{bmatrix} 1 & 1 & 3 & 2 \\ 2 & 3 & 6 & 1 \\ 2 & 2 & 6 & 4 \end{bmatrix} x = 0$

$\therefore \begin{bmatrix} 1 & 1 & 3 & 2 \\ 0 & 1 & 0 & -3 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = 0$

$\therefore$ For $\begin{bmatrix} x_1 \\ x_2 \\ 1 \\ 0 \end{bmatrix}$, $x_a = \begin{bmatrix} -3 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ -1 \\ 0 \end{bmatrix}$

For $\begin{bmatrix} x_1 \\ x_2 \\ 0 \\ 1 \end{bmatrix}$, $x_b = \begin{bmatrix} -5 \\ 3 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ -3 \\ 0 \\ -1 \end{bmatrix}$

$\therefore N = x = \begin{bmatrix} 3 & 5 \\ 0 & -3 \\ -1 & 0 \\ 0 & -1 \end{bmatrix}$

$\therefore$ Nullity $= 2$

Question 6: This is a bonus question for zero marks! Given

$$\mathbf{A} = \begin{bmatrix} 0 & \sqrt{3} & 0 & \sqrt{7} \\ 0 & 0 & \sqrt{15} & 0 \\ 0 & 0 & 0 & \sqrt{35} \\ 0 & 0 & 0 & 0 \end{bmatrix},$$

find the most general symmetric square matrix $\mathbf{B}$ such that $(\mathbf{BA})+(\mathbf{BA})^T = \mathbf{0}$ **(0 marks)**

(Total Marks: 40)

$$B = \begin{bmatrix} b_1 & b_2 & b_7 & b_4 \\ b_2 & b_5 & b_6 & b_3 \\ b_3 & b_6 & b_8 & b_9 \\ b_4 & b_7 & b_4 & b_{10} \end{bmatrix} \qquad \therefore B^T = \begin{bmatrix} b_1 & b_2 & b_3 & b_4 \\ b_2 & b_5 & b_6 & b_7 \\ b_7 & b_6 & b_8 & b_9 \\ b_4 & b_3 & b_9 & b_{10} \end{bmatrix}$$

$$\therefore BA = \begin{bmatrix} 0 & \sqrt{3}b_1 & \sqrt{15}b_2 & \sqrt{7}b_1 + \sqrt{35}b_7 \\ 0 & \sqrt{3}b_2 & \sqrt{15}b_5 & \sqrt{7}b_2 + \sqrt{35}b_6 \\ 0 & \sqrt{3}b_3 & \sqrt{15}b_6 & \sqrt{7}b_3 + \sqrt{35}b_8 \\ 0 & \sqrt{3}b_4 & \sqrt{15}b_7 & \sqrt{7}b_4 + \sqrt{35}b_9 \end{bmatrix}$$

$$\therefore (BA)^T = \begin{bmatrix} 0 & 0 & 0 & 0 \\ \sqrt{3}b_1 & \sqrt{3}b_2 & \sqrt{3}b_3 & \sqrt{3}b_4 \\ \sqrt{15}b_2 & \sqrt{15}b_5 & \sqrt{15}b_6 & \sqrt{15}b_7 \\ \sqrt{7}b_1+\sqrt{35}b_7 & \sqrt{7}b_2+\sqrt{35}b_6 & \sqrt{7}b_3+\sqrt{35}b_8 & \sqrt{7}b_4+\sqrt{35}b_9 \end{bmatrix}$$

$\because BA + (BA)^T = 0 \qquad \therefore b_1 = b_2 = b_7 = 0$

$\therefore \sqrt{3}b_3 + \sqrt{15}b_5 = 0$, $\sqrt{3}b_4 + \sqrt{35}b_6 = 0 \qquad \because b_6 = 0 \qquad \therefore b_4 = 0 \qquad \therefore b_9 = 0$

$\therefore \sqrt{7}b_3 + \sqrt{35}b_8 = 0 \qquad \therefore b_5 = -\frac{\sqrt{5}}{5}b_3 \qquad \therefore b_8 = -\frac{\sqrt{5}}{5}b_3$

$$\therefore B = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & -\sqrt{5} & 0 & 5 \\ 5 & 0 & -\sqrt{5} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$